Leandro Marsó

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Electronic Engineer who makes complex systems reliable, observable and secure. Twenty years across two industries: semiconductors, with six years at ClariPhy (now Marvell) implementing 100G and 200G optical communication ICs in 28 and 40 nm CMOS and building the embedded boards that control them; and cloud infrastructure since 2019, in platform engineering, cloud security and site reliability, much of it on carrier-grade voice (VoIP) platforms. What I bring: indicators and alerts people can trust; high-availability platforms; cloud security at scale; root causes found, from a timing path in silicon to a SIP response code in production; automation, now also with AI agents; and teaching it to others. Linux and networking since 2005.

Professional Experience

SIPEngines LLC — **Miami, FL (Remote)**
Infrastructure / SRE Engineer, Cloud and VoIP Observability — *July 2026 – Sept 2026*

- Defined service level indicators for SIP infrastructure by correlating response codes with call history, metrics and logs: a long-standing error rate fell tenfold and time to repair 20x; support now fixes issues before customers notice.
- Raised alert accuracy on one service tier from about 25% to 97% with a new indicator (Network Effectiveness).
- Made call quality, call setup delay, registration problems and per-carrier health visible across a multi-carrier voice platform, with dashboards kept as code.
- Recovered a production metrics data-loss incident, found its root cause and wrote the backfill procedure now used as standard practice.
- Use AI agents connected to the monitoring and SIP capture systems to cut toil and speed up investigations.
- Audited security and configuration drift across cloud accounts, removing over-permissive access; new hosts are deployed as code with zero-trust access only.
- Lowering compute costs by moving workloads to ARM instances; building a multi-tenant public status service.

Personal Projects — **marso.ar**
Platform Engineering and Applied AI Lab — *Ongoing*

- Run my own small production platform end to end: Kubernetes, GitOps, CI with test and security gates, metrics, logs and traces, dashboards as code, AI agents' cost tracked through an LLM gateway.
- Built Extractos, a public site that finds the exact moment a phrase is sung in a band's songs and plays that clip: vocal separation plus speech forced alignment, measured against hand-labelled ground truth (93% of lines within 0.3 s, up from 60% with the first approach).

Freelance — **US-based company (Remote)**
VoIP Platform Architect / DevOps Engineer — *Feb 2024 – July 2026*

- Designed and deployed a geo-redundant, high-availability, carrier-grade voice platform delivering a multi-tenant business phone service (PBX SaaS) to 10+ customers.
- Built its monitoring and alerting, with richer SIP call capture for faster call-flow debugging.

ASAPP — **Remote**
Cloud Security Engineer — *June 2021 – Aug 2023*

- Secured a large multi-cloud platform (AWS and GCP) at an AI-native company: thousands of Kubernetes pods, hundreds of hosts, and least-privilege access across 20+ cloud accounts used by hundreds of engineers.
- Owned the security monitoring platform (SIEM) and zero-trust privileged access to the infrastructure.
- Detected and contained sensitive data (PII, PCI) reaching the logs.
- Supported SOC 2 audits with dashboards and evidence for the auditors.
- Added vulnerability scanning to the delivery pipelines, built hardened machine images to the CIS Benchmarks, and cut storage costs within compliance retention rules.
- Took part in the on-call rotation for production and security alerts.

SiriusXM **Remote**
Platform Engineer *June 2019 – May 2021*

- Automated and optimized the deployments of the enterprise identity and single sign-on stack of a subscription business (40+ microservices), on premises and in the cloud.
- Took part in moving that stack to AWS, with the infrastructure provisioned as code.
- Built and maintained its delivery pipelines and configuration management.

Independent Contractor **Remote**
VoIP Platform Developer *Jan 2019 – June 2019*

- Built custom voice services for clients in Ecuador, Colombia, Israel and Argentina.
- Took a system to manage GSM gateways from idea to production in 3 weeks.
- Detected blocked or out-of-credit lines by audio fingerprinting; added call filtering, live monitoring and reports.

Universidad Provincial de Oficios Eva Perón **Villa Mercedes (San Luis)**
Project Leader - Full Stack Developer *Jul 2018 – Dec 2018*

- Led the deployment and customization of the university's enrollment and academic records system, coaching a developer team and managing contractors.

National University of San Luis, Engineering Faculty **Villa Mercedes (San Luis)**
Senior Teaching Assistant, Mathematics *Dec 2016 – Jun 2018*

- Taught mathematics to engineering students; moved the course material to version-controlled LaTeX. Meanwhile, completed the coursework of a Master's in Embedded Systems.

Instituto de Formación Docente Continua Villa Mercedes **Villa Mercedes (San Luis)**
GNU/Linux Systems Administrator *Oct 2015 – Dec 2016*

- Ran the servers behind the institution's management system; validated its major-version upgrade on a cloned test environment; added monitoring, security and automated backups.

Escuela Técnica Nº 17 V Brigada Aérea **Villa Reynolds (San Luis)**
High School Teacher *May 2015 – March 2016*

- Wrote three textbooks for the subjects I taught, published as open source on GitHub.

Servicios Tecnológicos Integrados (STI) **Córdoba (Córdoba)**
External Consultant and Trainer, IC Design *April 2015*

- Trained 15 engineers and PhDs in analog and digital integrated circuit design flows, using open-source tools.

ClariPhy Communications (now Marvell) **Córdoba (Córdoba)**
Senior Hardware and Software Developer, Embedded Systems *Aug 2013 – Oct 2014*

- Designed the hardware and software of data acquisition and control boards for 100 Gb/s and 200 Gb/s optical communication ICs.
- Designed the network protocol that controls the boards and implemented it twice: bare metal on an FPGA soft processor (Gigabit Ethernet) and on an ARM microcontroller (USB 3.0, RTOS).

ClariPhy Communications (now Marvell) **Córdoba (Córdoba)**
Senior IC Physical Design Engineer *Jan 2010 – Apr 2013*

- Implemented high-speed digital blocks of up to 2 million cells from RTL to GDSII in 28 nm and 40 nm CMOS.
- Closed timing and power across multiple operating scenarios: static timing and power analysis, multi-scenario constraints.
- Proposed hierarchy partitions for large designs together with the system designers.
- Automated the design flows on Linux compute grids.

ClariPhy Communications (now Marvell) **Córdoba (Córdoba)**
Senior IC Analog Layout Engineer *Oct 2008 – Dec 2010*

- Laid out high-speed (10 GHz) analog circuits in 40 nm CMOS, delivered free of LVS and DRC errors.

- Wrote parametric layout generators for the analog building blocks (differential pairs, current mirrors, loads, bypass capacitors) that are correct by construction, so new variants needed no manual rework.

Vanesa Duran S.A. **Córdoba (Córdoba)**
Network and GNU/Linux Administrator *Apr 2005 – Sep 2008*

- Ran the company network (Linux gateways, firewalls, a site-to-site VPN between two offices); virtualized critical legacy servers and set up business continuity and disaster recovery.
- Centralized logins and file sharing; cut hardware and energy costs by moving desktops to thin clients.

Skills

- Reliability and observability: service level indicators and alerts people trust; metrics, logs and traces end to end; incident response and root cause analysis; runbooks.
- Cloud platforms: AWS (5 years) and GCP; Kubernetes and containers; infrastructure as code; GitOps; delivery pipelines with test and security gates; cost optimization.
- Security: least-privilege access at scale, zero-trust access, security monitoring (SIEM), SOC 2 evidence, hardened images, vulnerability management.
- Real-time voice (VoIP): carrier-grade SIP platforms, high availability, call quality measurement, protocol-level troubleshooting.
- AI in operations: agents connected to monitoring and capture systems to investigate faster, with their usage and cost measured.
- Semiconductors and embedded: IC physical design (RTL to GDSII, timing and power closure, 28/40 nm CMOS), high-speed analog layout, parametric cell generation; board design, FPGA and ARM firmware, RTOS.
- Programming and automation: Python, Bash, C, Verilog, TCL, Perl.
- Teaching and leadership: trained engineers in IC design flows, taught university mathematics, led a development team, wrote three textbooks.
- Main tools: AWS, Kubernetes, Terraform/OpenTofu, Ansible, GitHub Actions, Argo CD, Prometheus, Grafana, Loki, OpenTelemetry, Kamailio, FreeSWITCH, Homer, Synopsys, Cadence.
- Languages: Spanish (native), English (fluent), Portuguese (fluent), Thai (conversational).

Education

Master's in Embedded Systems, thesis pending (Universidad Nacional de San Luis) 2016 – 2018
Electronic Engineer (Universidad Nacional de Córdoba) 1998 – 2015

Courses

- Advanced Programming of Embedded Systems (2017): PhD Mario M. Berón. Facultad de Ciencias Físico Matemáticas y Naturales - Universidad Nacional de San Luis. 60 hours approved course.
- Programmable Logic (2017): MSc Cristian Cisterna. Facultad de Ciencias Físico Matemáticas y Naturales - Universidad Nacional de San Luis. 60 hours approved course.
- Distributed Systems (2017): MSc Roberto Kiessling. Facultad de Ciencias Físico Matemáticas y Naturales - Universidad Nacional de San Luis. 60 hours approved course.
- Architecture and Programming of Embedded Systems (2017): PhD Mario M. Berón. Facultad de Ciencias Físico Matemáticas y Naturales - Universidad Nacional de San Luis. 60 hours approved course.
- Data Mining (2016): PhD Marcelo Luis Errecalde. Facultad de Ciencias Físico Matemáticas y Naturales - Universidad Nacional de San Luis. 60 hours approved course (Resolution RR-3-383/17).
- Quantitative Data Analysis (2016): MSc Jorge Leporatti and MSc Fabricio Penna. Facultad de Ingeniería y Ciencias Agropecuarias - Universidad Nacional de San Luis. 40 hours approved course (RR-1-1157/16)